Redundant Experiments: Independent Re-Run Comparison

This document compares the results in `paper.tex` (run by Diego) against an independent re-run of the same eval sweep (results supplied by Riyasat, unpacked under `evals/riyasat/`), using the exact same plots and table layout as the main paper for direct comparison. **Caveat:** Riyasat's export did not include OLMoE's *Working-set reward*, *Option-critic controller*, or *MELINOE* runs – those three rows below are copied in from Diego's own results (not an independent re-run) purely to keep the table complete; do not read them as a redundancy check. Every other row/figure here *is* an independent re-run and should be compared against the corresponding number in `paper.tex`.

1 Downstream Task and Working-Set Metrics

Table 1: Phi-tiny-MoE-instruct, redundant run (1024 held-out prompts, $c = 4$, top-2). Base row is absolute; other rows are Δ from this run's own base model. Compare against Table 1 of `paper.tex`.

Objective	lm-eval-harness score (Δ from base) MMLU	MMMLU	GSM8K	HumanEval	MATH	working-set efficiency Hit ratio	Misses	ECI
Base (untuned)	0.619	0.344	0.624	0.031	0.085	0.570	648.6	0.891
Causal LM alone (SFT)	+0.003	−0.018	+0.017	0.000	−0.006	0.568	789.3	0.892
Working-set reward	−0.023	+0.011	−0.055	+0.012	−0.011	0.621	437.6	0.888
Working-set reward + SFT	−0.004	+0.006	−0.002	−0.006	−0.006	**0.789**	**401.6**	0.691
Option-critic controller	−0.024	−0.020	−0.110	−0.012	−0.076	**1.000**	**0.1**	0.279
MELINOE	0.000	−0.010	−0.138	−0.019	−0.069	0.879	239.0	0.674

Table 2: OLMoE-1B-7B, redundant run where available (1024 held-out prompts, $c = 16$, top-8). Base and SFT rows are an independent re-run; the last three rows are carried over from `paper.tex` (see caveat above). Compare against Table 2 of `paper.tex`.

Objective	lm-eval-harness score (Δ from base) MMLU	MMMLU	GSM8K	HumanEval	MATH	working-set efficiency Hit ratio	Misses	ECI
Base (untuned)	0.565	0.377	0.680	0.311	0.056	0.686	1879.5	0.895
Causal LM alone (SFT)	−0.021	−0.014	−0.095	−0.201	−0.012	0.678	1957.9	0.897
– rows below are carried over from Diego's run, not an independent re-run –								
Working-set reward	−0.001	−0.010	−0.007	+0.018	+0.004	0.759	1004.2	0.895
Option-critic controller	−0.038	−0.066	−0.138	−0.171	−0.054	0.684	2012.7	0.888
MELINOE	−0.073	−0.055	−0.245	−0.232	−0.054	0.763	1541.9	0.838

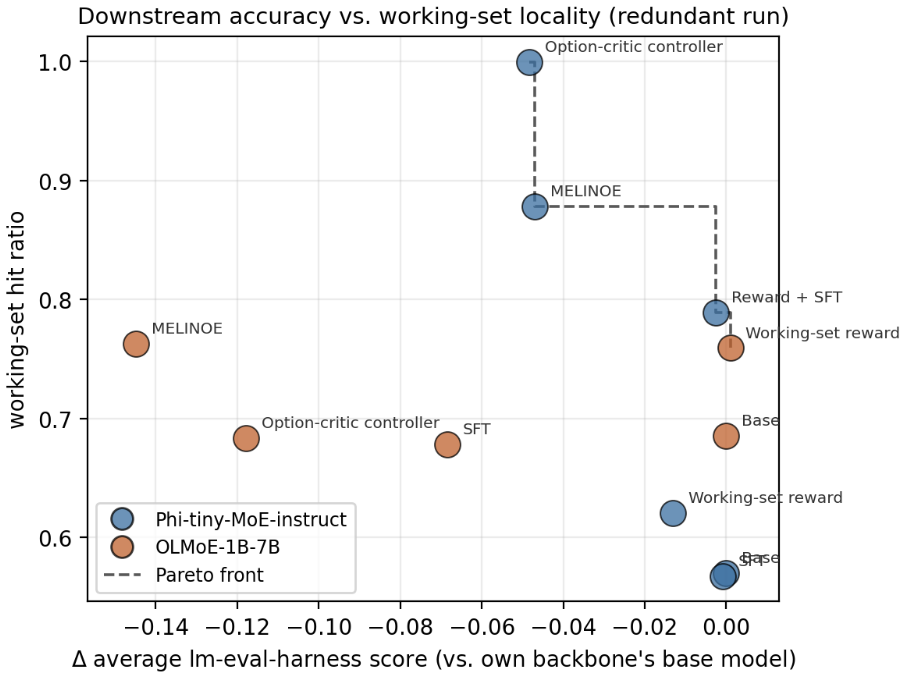

Figure 1: Redundant run's version of Figure 1 in `paper.tex`: change in average lm-eval-harness score versus working-set hit ratio, relative to each backbone's own base model in *this* run. Compare the shape of the Pareto front and the relative position of each labeled point against the main paper's figure.

2 Downstream Accuracy vs. Working-Set Locality

3 Prompt-Conditioned Capacity Sweep

Riyasat's export included the full prompt-conditioned cache-size sweep for *both* backbones (Phi-tiny-MoE at $c \in \{2, 4, 8\}$ and OLMoE at $c \in \{8, 16, 32\}$), unlike the main paper which only had OLMoE's sweep available at submission time. The two capacity ranges are put on a common x-axis as % of each backbone's total expert count, so both curves can be read on one plot (Phi: 16 experts, OLMoE: 64 experts – $c = 2, 4, 8$ and $c = 8, 16, 32$ land on the same 12.5%/25%/50% marks).

Table 3: Prompt-conditioned policy, redundant run, swept over capacity.

Backbone	Capacity c	Hit ratio	Tokens/s (base-normalized)
Phi-tiny-MoE-instruct	2	0.651	181.0
Phi-tiny-MoE-instruct	4	0.831	184.4
Phi-tiny-MoE-instruct	8	**0.944**	**186.3**
OLMoE-1B-7B	8	0.508	259.4
OLMoE-1B-7B	16	0.741	272.3
OLMoE-1B-7B	32	**0.903**	**279.5**

4 Expert Routing Traces (Redundant Run)

Routing traces for the objectives that were independently re-run. OLMoE's working-set reward, option-critic controller, and MELINOE traces are omitted here since those rows are carried over from Diego's run

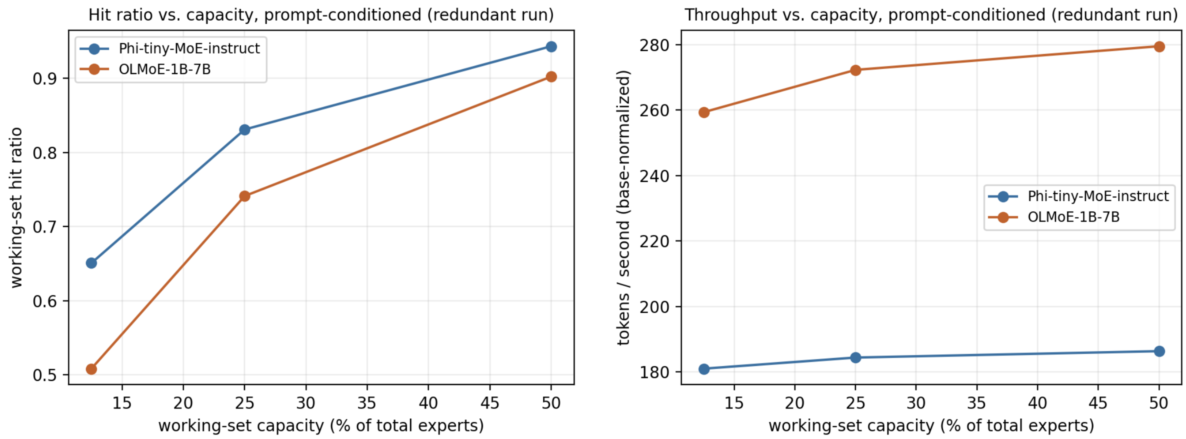

Figure 2: Prompt-conditioned hit ratio (left) and base-normalized tokens/second (right) vs. working-set capacity, as a fraction of each backbone's total expert count, for the redundant run. Compare against `paper.tex`'s OLMoE-only scaling figures (Appendix figures for the main paper) – this version additionally has the Phi-tiny-MoE curve, not available from Diego's own sweep at submission time.

(identical by construction, not a useful comparison point).

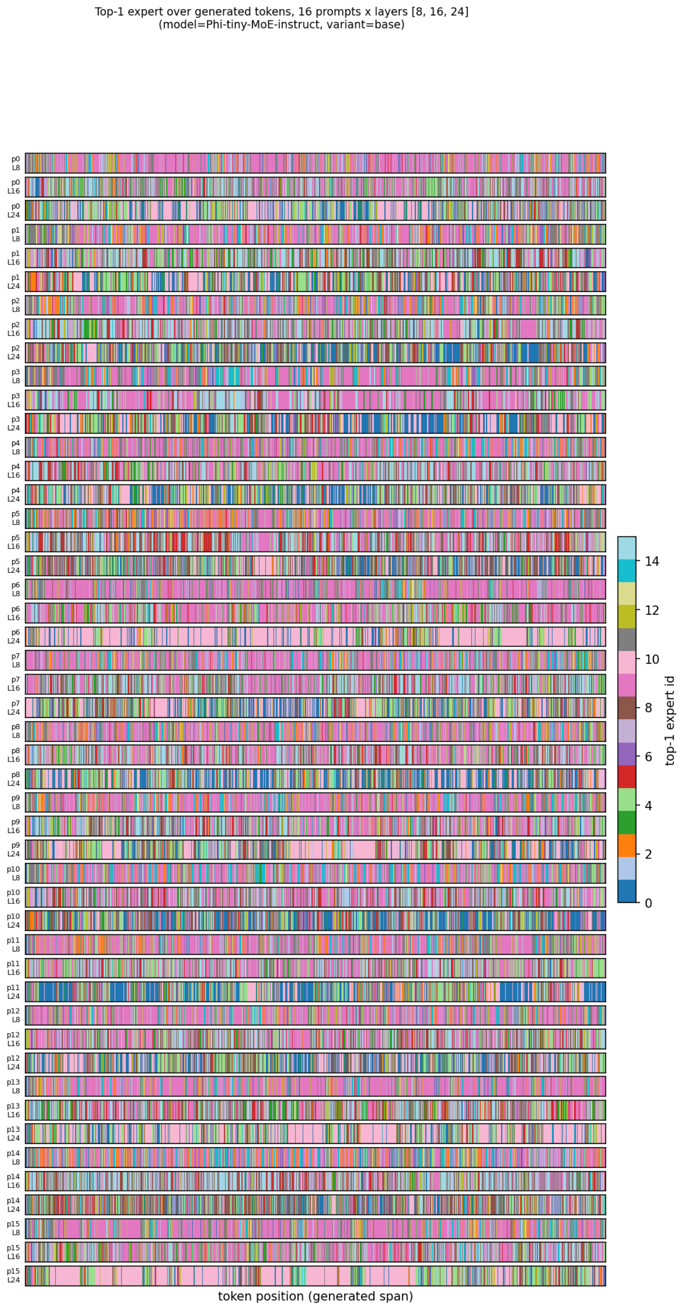

Figure 3: Phi-tiny-MoE-instruct, base (untuned), redundant run.

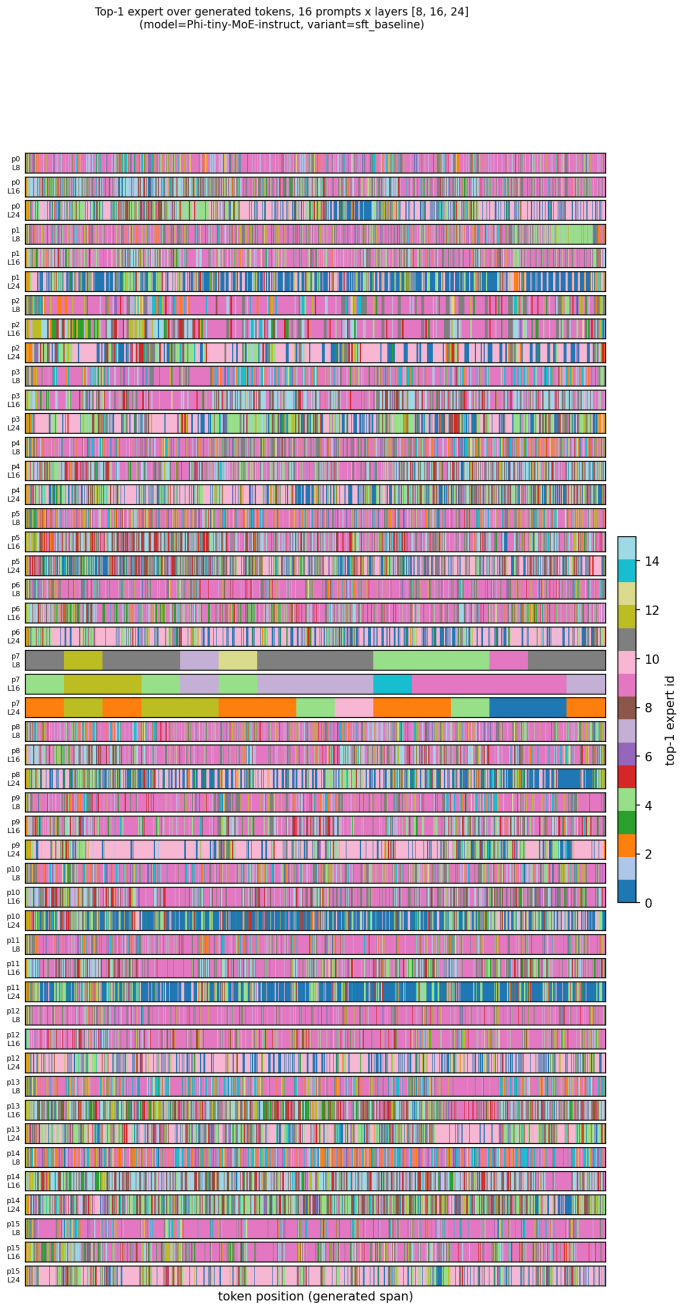

Figure 4: Phi-tiny-MoE-instruct, causal LM alone (SFT), redundant run.

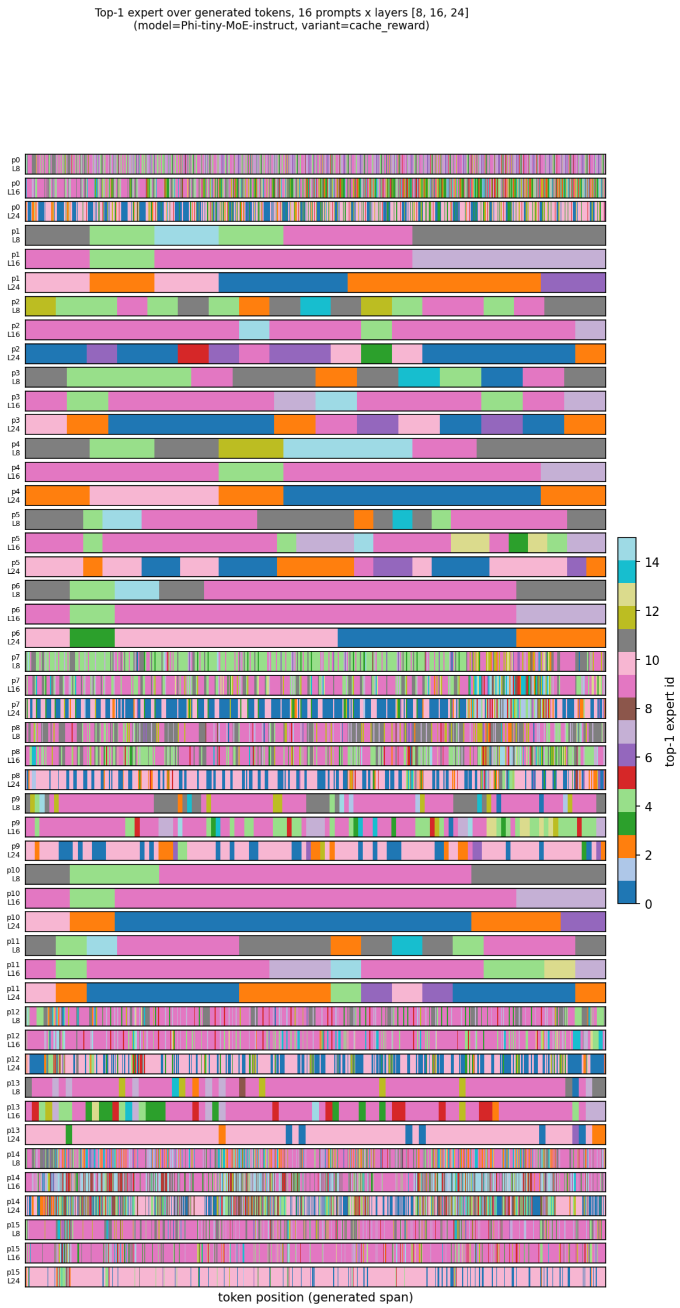

Figure 5: Phi-tiny-MoE-instruct, working-set reward alone, redundant run.

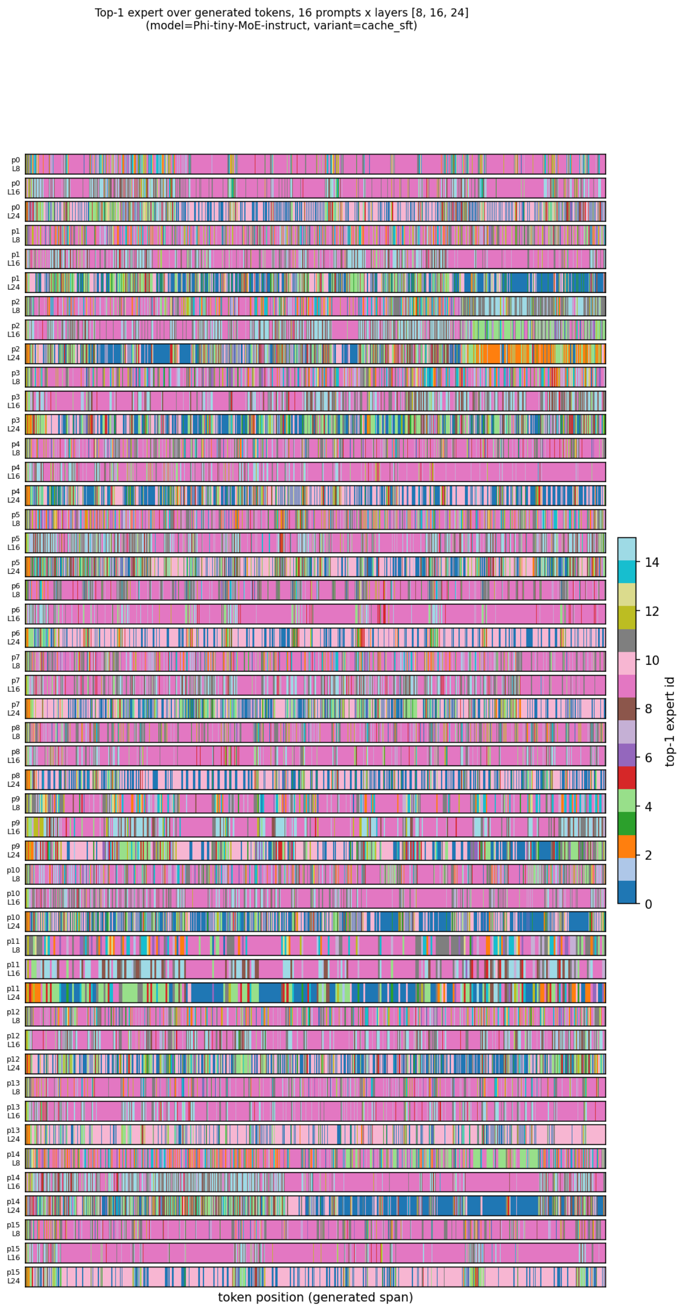

Figure 6: Phi-tiny-MoE-instruct, working-set reward + SFT, redundant run.

Figure 7: Phi-tiny-MoE-instruct, option-critic controller, redundant run.

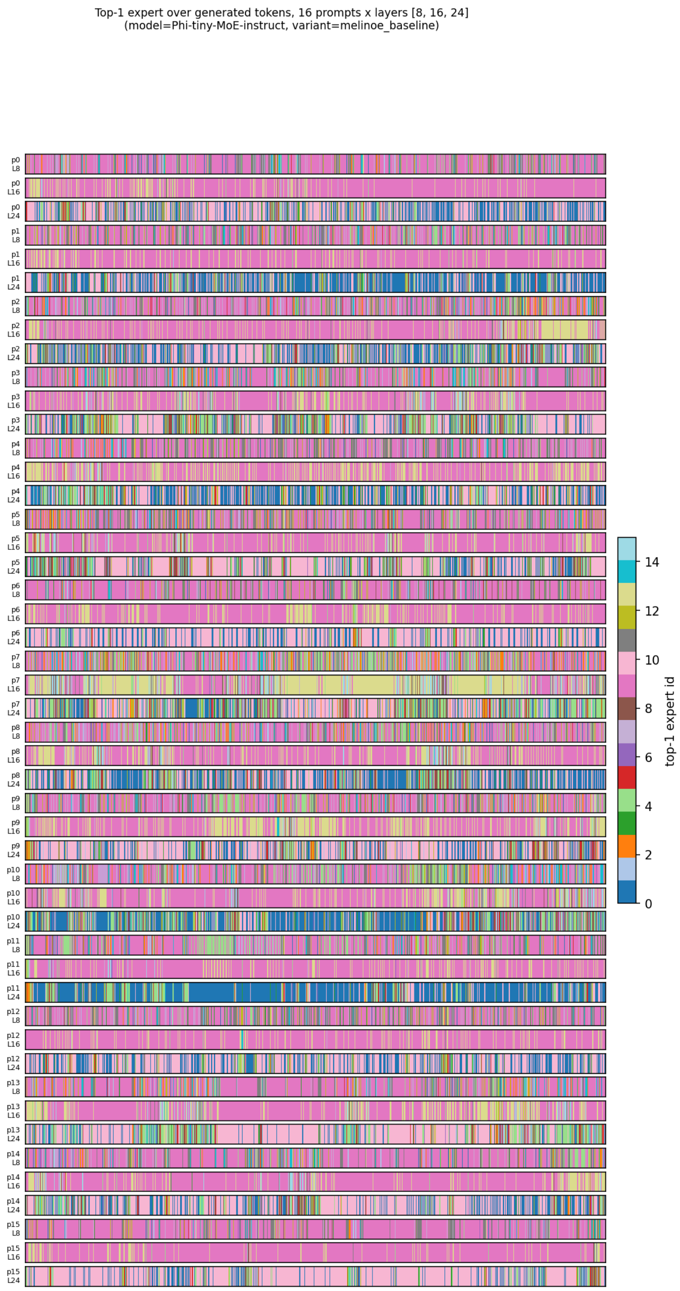

Figure 8: Phi-tiny-MoE-instruct, MELINOE, redundant run.

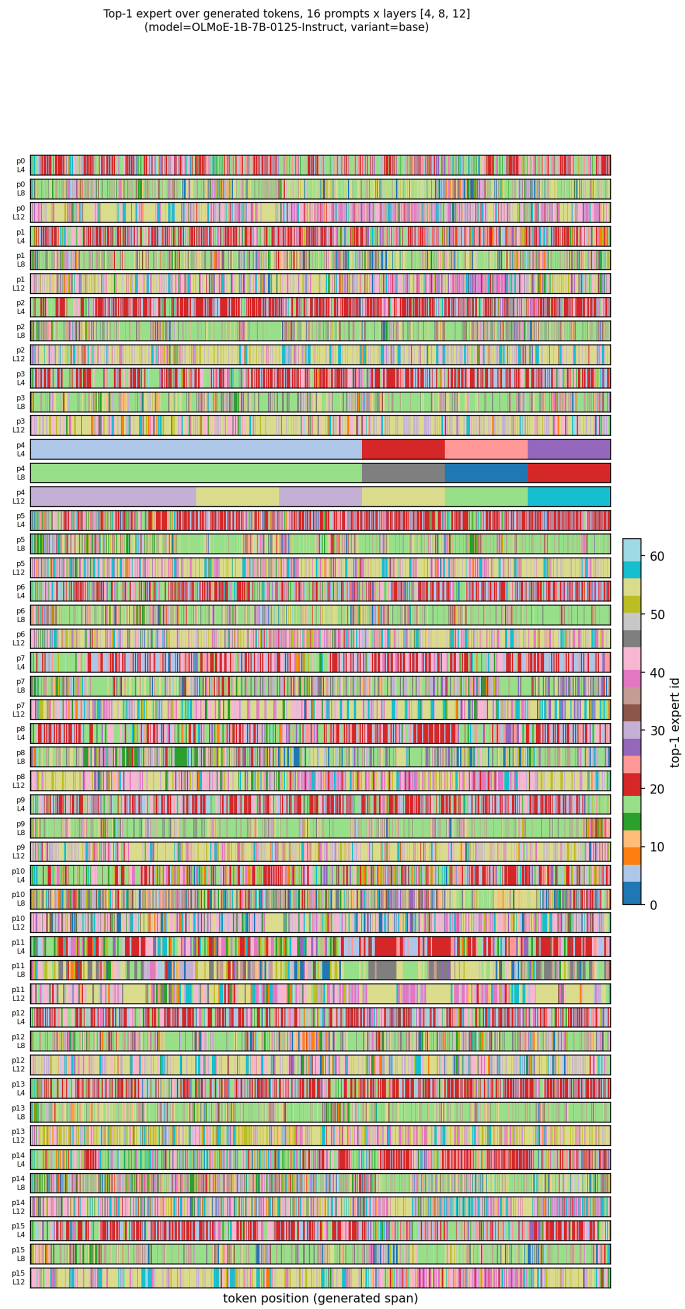

Figure 9: OLMoE-1B-7B, base (untuned), redundant run.

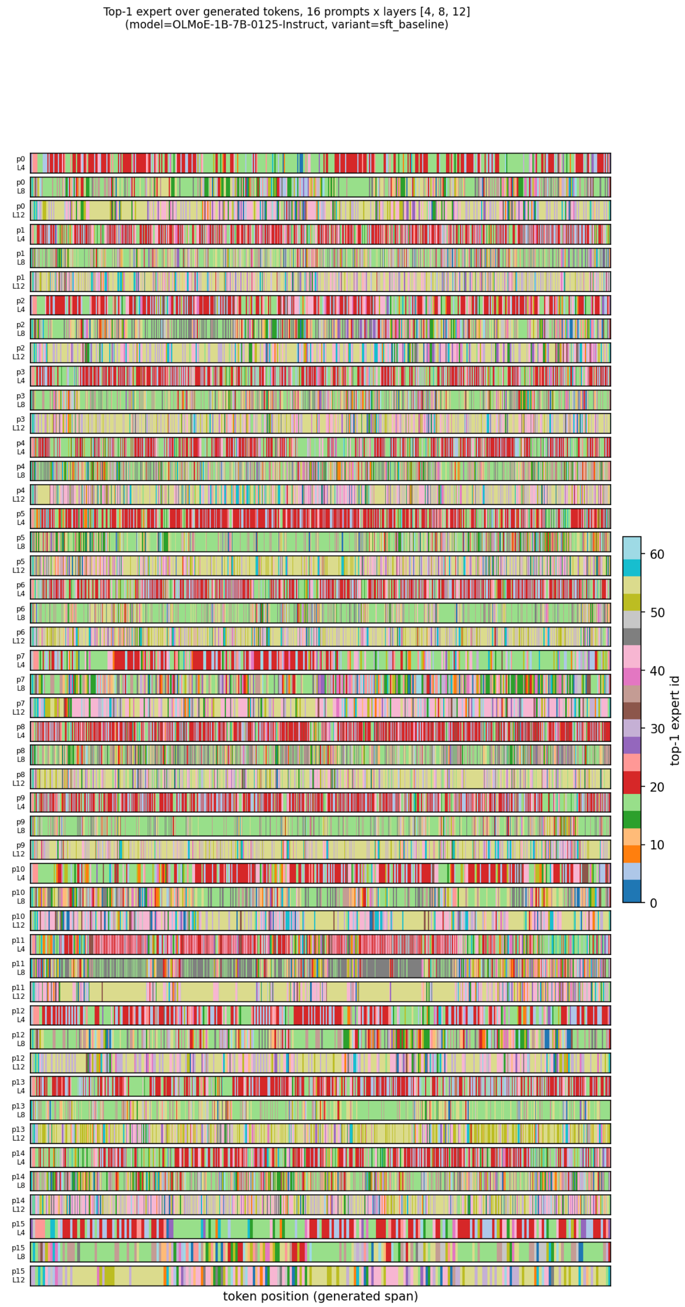

Figure 10: OLMoE-1B-7B, causal LM alone (SFT), redundant run.